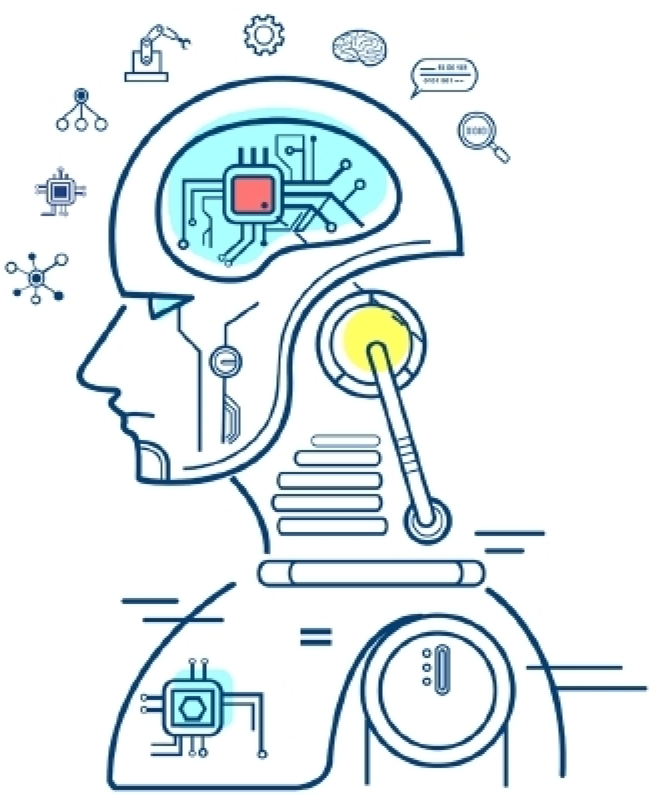
Top Open Source Tools *for* Artificial Intelligence and Machine Learning

'Data is more valuable than gold'. This is the mantra of modern computing. Enormous amounts of data are being generated every minute throughout the world. The entry of AI and ML has facilitated the processing of this data and its use in the enterprise as well as in various other fields. Here is a bird's eye view of trending open source tools for AI and ML.



2018 can well be remembered as the year where data first demonstrated its dominance, with a visible impact not only on science and technology but also on global politics and socioeconomic conflict, especially in developing nations. While we witnessed the trouble it can foment, we were made painfully aware of the terrible costs incurred if these tools are used unethically. On the whole, this article seeks to adopt an objective view as we look at how artificial intelligence (AI) is maturing, backed by large scale research efforts across the world from Silicon Valley in the West to China in the East.

Top machine learning (ML) frameworks

The stalwart across the field remains Google's TensorFlow that provides an enterprise-grade system to train, test and deploy deep neural networks at scale. It has steadily grown and is supported by an ecosystem of visualisation, data manipulation and interpretability tools that make it a ubiquitous solution when it comes to scalable machine learning. With the added support of Keras integration, Google is now trying to shorten the learning period for developers to work with TensorFlow.

Last year we saw the emergence of PyTorch as one of the frameworks preferred by machine learning researchers who often chose not to use the dominant TensorFlow, given the flexibility and features in the younger, lightweight, open source, deep learning library supported by and extensively used by Facebook. Most comparisons of state-of-art frameworks are focused on TensorFlow and PyTorch, arguably given their strong adoption rate in academia and industry, shadowing the others like Caffe, Theano and Microsoft's Cognitive Toolkit (CNTK). Following these, there's also the Apache MXnet project with the Gluon interface, which seeks to provide simple and quick building blocks that allow users to speedily prototype deep learning models.

Scikit-learn remains a widely used open source framework to prototype and deploy classifiers for machine learning, but is more focused on providing a 'workbench' in order to avoid the boilerplate code that presents a challenge for picking up frameworks like TensorFlow and PyTorch.

We do have Spark MLib and CNTK in use across enterprises. Netron, a popular visualisation library for neural networks, now also supports CNTK while Spark MLib is seeing steady adoption as companies start out with building scalable data streaming pipelines. In combination with Mahout and Apache's other products for Big Data management and architecture, Apache has released SystemML as an addition to its repertoire of open source tools at the intersection of Big Data and machine learning.

Libraries such as Fast.ai's recently released software have advanced the state-of-the-art in some disciplines within natural language processing. Edward has been released as a probabilistic programming toolkit built atop TensorFlow (soon to be integrated within it), while Lime is another library supporting greater interpretability for deep neural networks. All these are seeing increased use as issues of privacy, ethics, and understanding of biases in data acquire greater importance within the industry. Many traditional applications also rely on machine learning capabilities in Java and R via frameworks such as deeplearning4j. Overall, the AI and ML space is bustling with developments that one needs to follow, and change seems to be the only constant.

Top tools for artificial intelligence in the cloud

'Artificial Intelligence-as-a-Service' is trending, especially because small-scale companies do not wish to do the heavy lifting of setting up end-to-end data pipelines, but would